Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/03/2006
PI Name:	Squire, Maria Ph.D.		
Project Title:	Identification of Chromosomal Regions Responsible for Loss of Bone Strength in Hindlimb Unloading		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Scranton	**State:**	PA
Zip Code:	18510	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2004	**End Date:**	06/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The loss of bone and skeletal muscle in astronauts can be severe, but is heavily influenced by genetics. Our studies using genetically distinct female BALB/cByJ and C3H/HeJ mice have shown strain-specific skeletal sensitivities to mechanical unweighting. The overall hypothesis of this proposal is that susceptibility to disuse- induced changes in the musculo- skeletal system is genotype- dependent. Understanding the genetic variability and pathways guiding disuse-induced changes in the musculo-skeletal system will provide additional steps towards identifying appropriate screening tools and treatment regimens for astronauts most at risk for suffering from osteopenia and sarcopenia during prolonged space travel.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated:)